



Q1. What is barcode? Can the barcode scanner be affected by light and radio frequency waves?

Omar

A1. A barcode is a series of parallel black bars and white spaces, both of varying widths. Each combination or sequence of bars and spaces is a code representing information such as product type, size, manufacturer, or origin of shipment. The barcode structure is shown in Fig. 1.

Barcodes usually come in different sizes. The smaller the size of the barcode, the less the number of characters it can encode. 1D barcode, also called linear barcode, can hold around 8 to 25 characters.

The bar patterns representing the data (numerical characters, alphabet, etc) are arranged from left to right. The barcode length defines the total length including the right and left quiet zones. The barcode height recommended is as high as the printer permits. It is recommended to keep the height greater than 15 percent of the barcode length.

A barcode scanner uses an LED or laser as light source to capture the image of the barcode, and the decoder decodes and passes it on to a computer in traditional data format.

A study has found that rapid fluctuations in light source seriously interfere with the barcode scanner, thereby degrad-

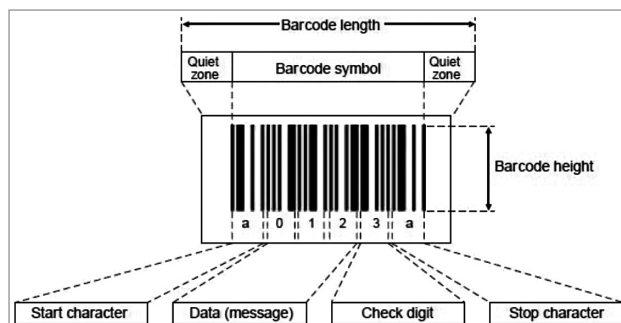


Fig. 1: Barcode structure (Credit:www.keyence.com)

ing the reading performance. The interference typically generated in the LED driver by a switched mode power supply (SMPS) has frequencies up to several MHz that can seriously affect the barcode reading performance. However, so far there is no clear evidence or study that supports the external radio frequency signals affecting the performance of a barcode scanner.

Q2. Which solder flux and solder wire are recommended for use in a general-purpose PCB?

Ellyse Perry

A2. Flux is a water based paste that contains a cleaning agent, activator, and wetting agent used in soldering to remove oxides from the area to be soldered. A good flux cleans the area to be soldered and allows the melted solder to flow and bond with the metal surfaces. The activator is a specific chemical designed to remove oxides, oil, and dirt.

Commonly there are three types of fluxes for electronics applications: rosin, water soluble, and no-clean fluxes. These fluxes are selected depending on the need and applications.

Fluxes can be made from ammonium chloride, hydrochloric acid, zinc chloride, or rosin. Water-soluble paste fluxes are applied by brush or dispensed from a syringe into the appropriate area, then heated using conventional methods. A no-clean flux does not require any cleaning after soldering. It is preferable to use the flux that matches the solder wire you are using.

There are three main categories of solder wire: Lead-based, lead-free, and flux-core solder.

Lead based is a type of traditional solder wire with 60/40

(tin/lead) mixture, the most common one available in the market. Its melting point is around 180-190 degrees Celsius. With higher tin concentration, it will have better tensile and shear strengths. You can also find non-corrosive solder wire having 63% tin and 37% lead mixture with good solderability and insulation resistance.

The second type is a lead-free solder wire. Lead affects health and environment and so lead based solder wire is not the ideal solution. Lead-free solder usually contains 99.3% tin and 0.7% copper with higher melting temperature of around 217 degrees Celsius.

The third is solder wire having the flux embedded in the solder wire itself. It does not require additional flux while soldering.

Solder flux from Noel and Glare India are found good enough for hobbyists. Bharti, Amico, TINNZTES, Globomotive, and Kester are some other popular solder wire brands. These brands are being mentioned based on our personal experience and reviews found online and are not necessarily the best ones.